Roll No: _____

Thapar Institute of Engineering and Technology
Department of Computer Science and Engineering
Auxiliary Examination

BE CSE Third Year
Course Name: Quantum Computing
Time: 3 hours (10:00 am-1:00 pm)
Marks : 100

Course Code: UCS619
Date: February 24, 2024
Name of Faculty: Dr Samya Muhuri

Note: Attempt all questions with proper justification and working. State your assumptions (if any) and show your working clearly.

1. (a) A system in the state

$$|\psi\rangle = \frac{1}{\sqrt{8}}|00\rangle + \sqrt{\frac{3}{8}}|01\rangle + \frac{1}{2}|10\rangle + \frac{1}{2}|11\rangle$$

What is the probability that measurement finds system in the state $\phi = |10\rangle$? Find using Born rule. **(6)**

(b) What is the probability that measurement finds the first qubit in the state $|0\rangle$? What is the state of the system after measurement? **(4)**

2. (a) A qubit is in the state

$$|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$$

Using the projection operators P_0 and P_1, indicate the probability of finding $|0\rangle$ and the probability of finding $|1\rangle$ when a measurement is made. **(6)**

(b) Using the appropriate assumptions, describe any two postulates of Quantum Mechanics in a detailed manner. **(4)**

3. (a) Differentiate between Teleportation and Superdense coding. Explain the procedure of Superdense coding assuming that the third party generates the entangled pair $\frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$. **(6)**

(b) Proof No cloning theorem with any one method. **(4)**

4. (a) Using the matrix representation of the Hadamard gate, write down the matrix $(H \otimes H)(|0\rangle \otimes |1\rangle)$. Show that this is equivalent to

$$|\psi\rangle = (\frac{|0\rangle + |1\rangle}{\sqrt{2}})(\frac{|0\rangle - |1\rangle}{\sqrt{2}})$$

(5)

(b) The standard way to write the state of a qubit using the bloch sphere is

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle$$

show that on the y-axis of the Bloch Sphere $y+ = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$ **(5)**

5. (a) Alice creates a 8-bit string
$|0\rangle\,|1\rangle\,|+\rangle\,|0\rangle\,|0\rangle\,|-\rangle\,|+\rangle\,|-\rangle$
If Bob randomly measures in the following order using the bases
$\{|0\rangle, |1\rangle\}\{|\pm\rangle\}\{|0\rangle, |1\rangle\}\{|\pm\rangle\}\{|0\rangle, |1\rangle\}\{|0\rangle, |1\rangle\}\{|\pm\rangle\}\{|\pm\rangle\}$
describe the bit string that Alice and Bob keep using BB84 algorithm.
(7)

(b) What is the fundamental difference between classical and quantum cryptography? **(3)**

6. (a) Prove that $X = HZH$ using matrix multiplication method. **(5)**

(b) Show that the Fredkin gate act as AND Gate. **(5)**

7. (a) Define Quantum Fourier Transform (QFT) for an n-qubit quantum system. **(5)**

(b) Define Bell's inequality with respect to EPR paradox. **(5)**

8. Using CCNOT and CNOT gate, design the quantum circuit for half adder and write down the QASM or Python code for the same. **(10)**

9. Consider a function with two inputs such that $f(x) = 1$. Explicitly show that the Deutsch-Jozsa algorithm works in this case by generating the vector $|y\rangle = |00\rangle$ as the final output.
(10)

10. Based on the Peter Shor's 9-qubit-code, show the process of rectification of one qubit X-bit error which happened in 5th qubit.
(10)